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AI-Optimized Science Metadata Format

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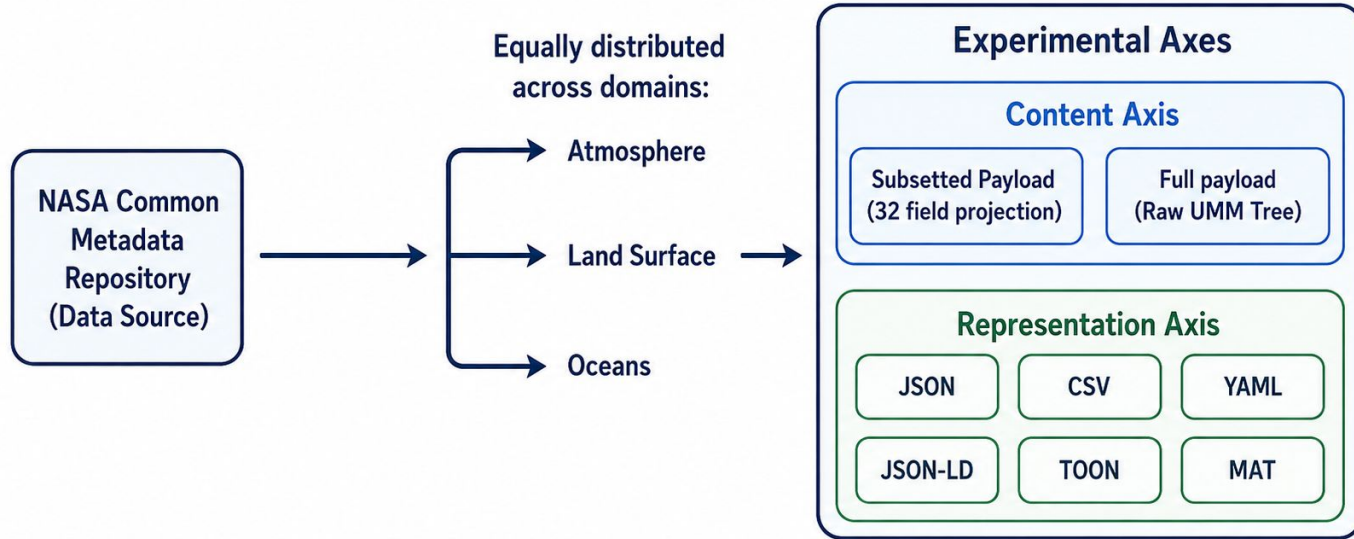
Research Questions

- How do Large Language Models (LLMs) interact with and utilize domain-specific metadata specifically in complex archives like NASA's Common Metadata Repository (CMR)?
- How can metadata architectures be optimized to improve AI search, retrieval, and accuracy?

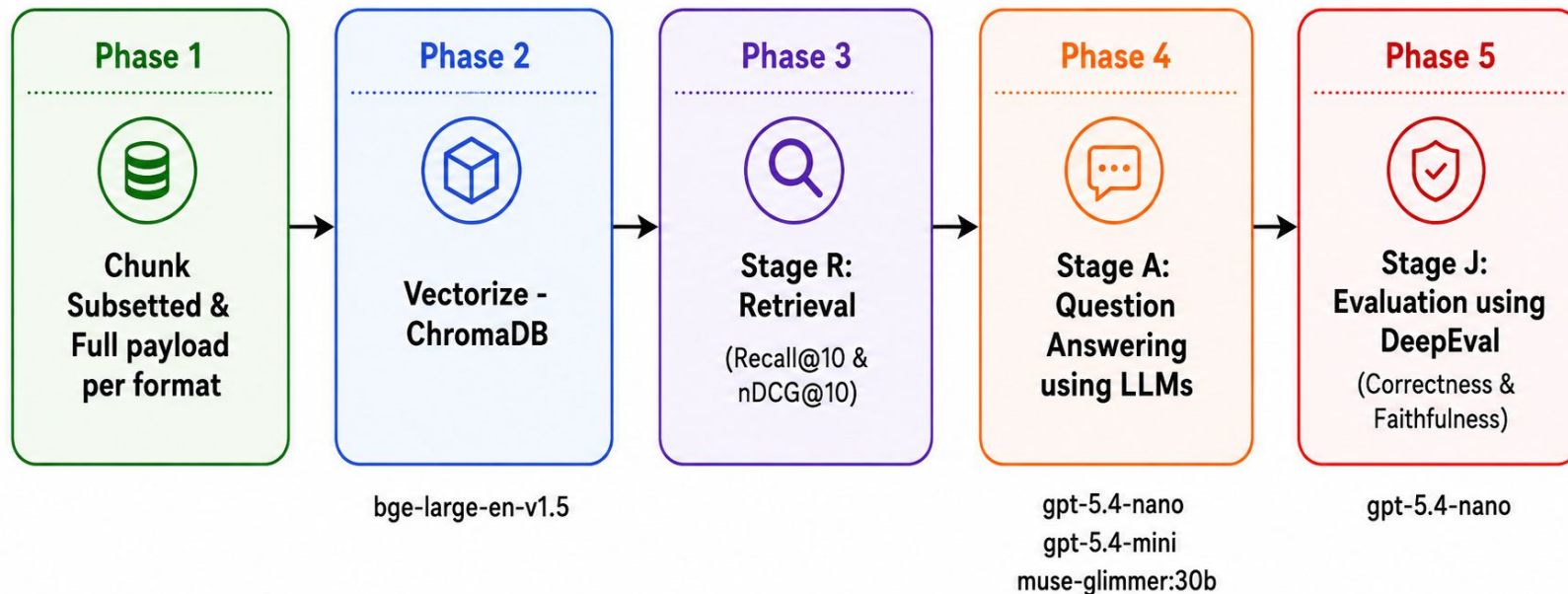
Hypothesis

- H1: Content dominates - field selection matters more than metadata format.
- H2: Format effect is small at equal content.
- H3: The axes (retrieval, reasoning, cost) rank formats differently.
- H4: The best metadata format varies by model family.

Data Gathering and Setup for Experiments



Experimental Workflow



Experiment 1 - Token Utilization

- Goal: Measure the token cost of serialization format at constant content.

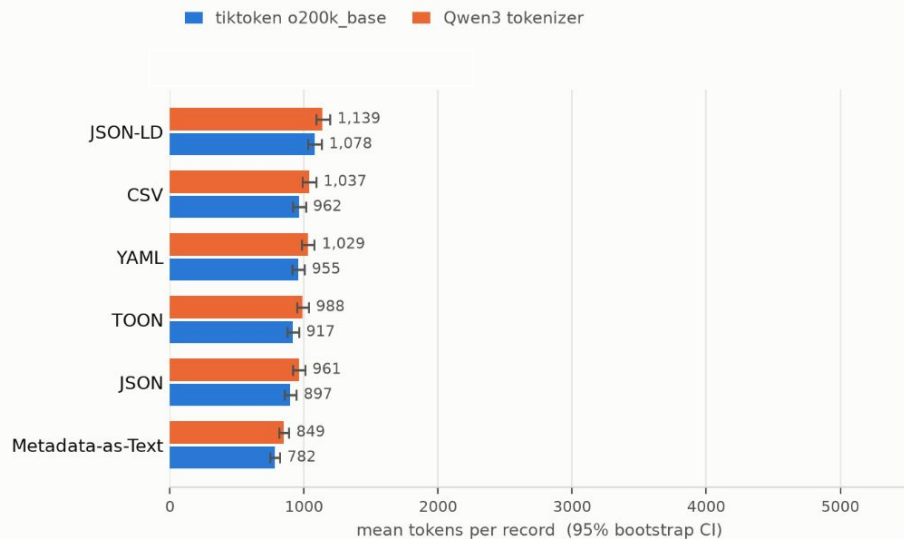


Fig 1: Mean token counts per record against subsetted records

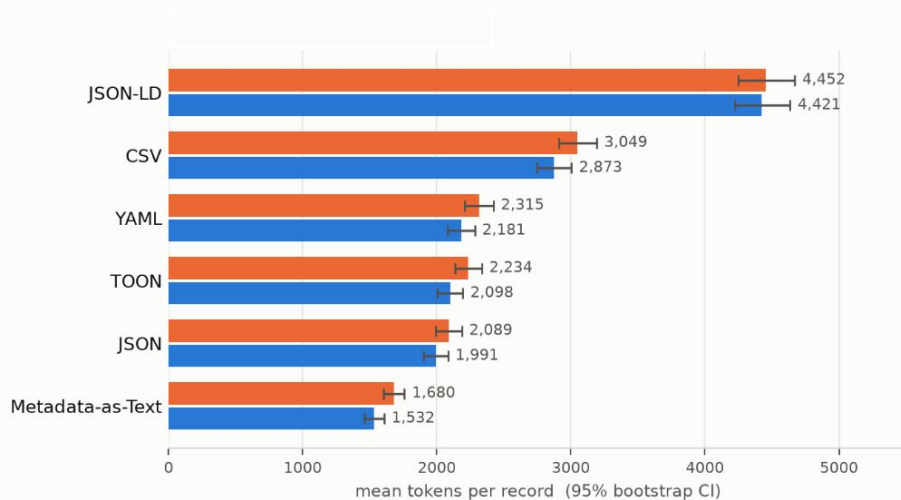


Fig 2: Mean token counts per record against full records

Experiment 2 - Retrieval: Faceted vs Unfaceted

- Goal: Measure record retrieval quality as a function of format under a fixed data embedding pipeline.

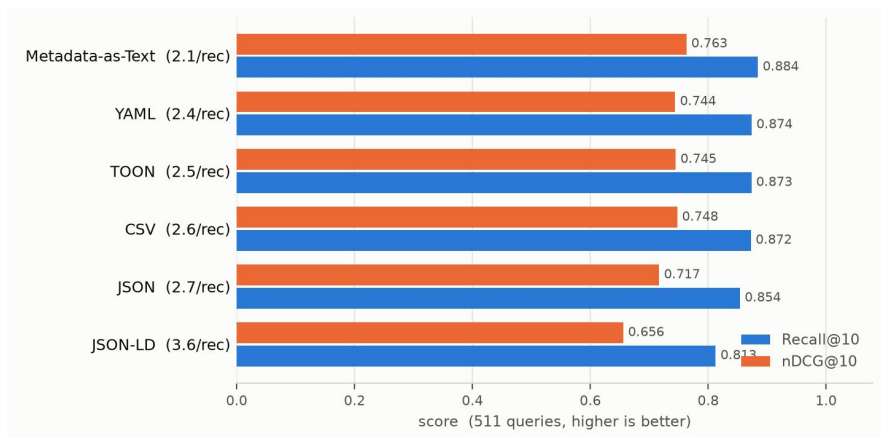


Fig 1: Recall@10 and nDCG@10 against subsetted records

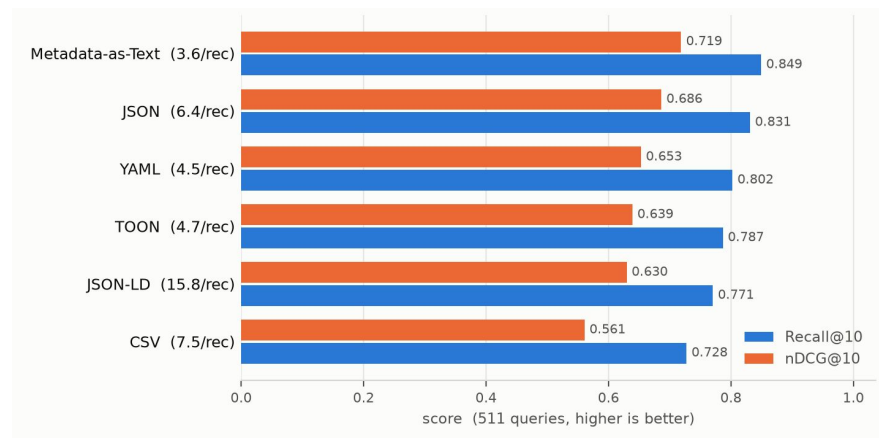


Fig 2: Recall@10 and nDCG@10 against full records

Experiment 3 - Retrieval - QA - Evaluation

- Goal: Measure end-to-end answer quality.

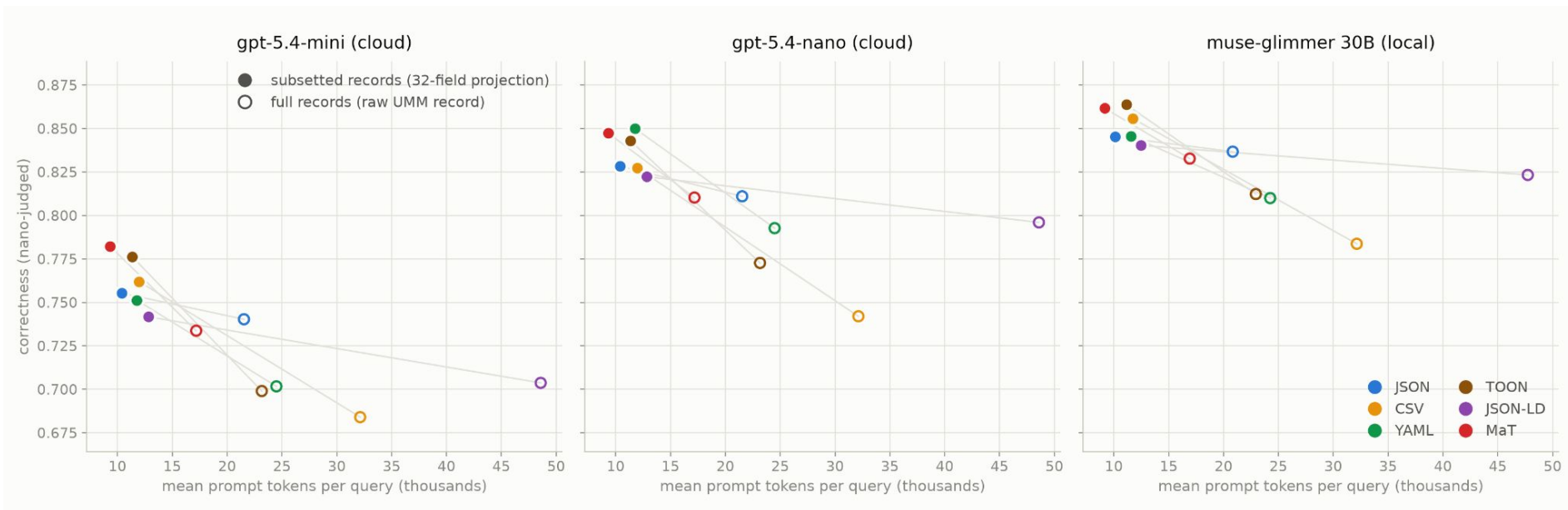


Fig 1: Correctness vs Mean Prompt tokens per query (thousands)

Experiment 3 - Retrieval - QA - Evaluation

- Goal: Measure end-to-end answer quality.

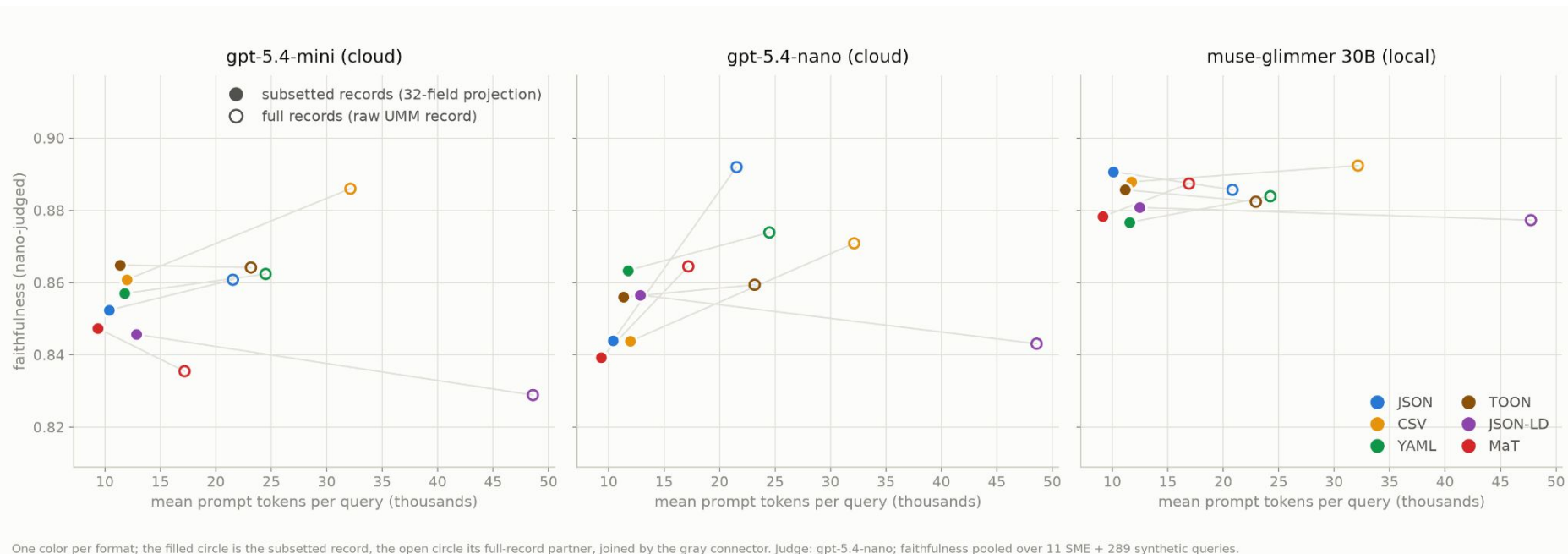


Fig 1: Faithfulness vs Mean Prompt tokens per query (thousands)

Experiment 4 - Enhanced Retrieval

- Goal: Measure whether an LLM-generated summary appended to each record improves retrieval.

Format	R@10 before	with summary	Δ	relative
JSON-LD	0.813	0.878	+0.066	+8.1%
CSV	0.872	0.911	+0.039	+4.5%
TOON	0.873	0.911	+0.038	+4.4%
JSON	0.854	0.887	+0.034	+4.0%
YAML	0.874	0.905	+0.031	+3.5%
MaT	0.884	0.903	+0.019	+2.1%

Fig 1: Recall@10 improvement against faceted records

Format	R@10 before	with summary	Δ	relative
CSV	0.728	0.803	+0.076	+10.4%
JSON-LD	0.771	0.847	+0.076	+9.9%
JSON	0.831	0.874	+0.043	+5.2%
TOON	0.787	0.828	+0.041	+5.2%
YAML	0.802	0.838	+0.035	+4.4%
MaT	0.849	0.883	+0.033	+3.9%

Fig 2: Recall@10 improvement against unfaceted records

Experiment 5 - Ablation Study for Measuring Recall

- Goal: Measure how Recall@10 compares to baseline when we drop the abstract, replace it with AI summary or append AI summary to the record.

Recall@10 (relative Δ vs baseline in parentheses)

	baseline	dropped	replaced	appended
Subsetted records	0.862	0.795 (-7.7%)	0.853 (-1.0%)	0.899 (+4.4%)
Full records	0.795	0.675 (-15.0%)	0.834 (+4.9%)	0.845 (+6.4%)

nDCG@10 (relative Δ vs baseline in parentheses)

	baseline	dropped	replaced	appended
Subsetted records	0.729	0.647 (-11.2%)	0.698 (-4.2%)	0.759 (+4.1%)
Full records	0.648	0.526 (-18.8%)	0.680 (+4.9%)	0.701 (+8.2%)

Fig 1: Recall@10 and nDCG@10 averages across all formats

Study Summary

- H1: Content dominates
 - Choosing what goes in context moves correctness $1.7\times$ more than picking the best format while reducing tokens by half.
- H2: Format effect is small at equal content.
 - Best vs worst format at equal content is 3-5% per LLM model.
- H3: The axes (retrieval, reasoning, cost) rank formats differently.
 - No format wins all three axes.
- H4: Best format varies by model family.
 - Three different format winners on three LLM models but the spread is within error margins.

Research Questions - Answered

- How do Large Language Models (LLMs) interact with and utilize domain-specific metadata specifically in complex archives like NASA's Common Metadata Repository (CMR)?
 - High quality metadata improves performance more than any one format.
- How can metadata architectures be optimized to improve AI search, retrieval, and accuracy?
 - Curate content first; format is a second-order choice.
 - A well written abstract/summary of the record can improve retrieval efficiency.
 - Metadata-as-Text can provide better results at a reduced cost/token.

Conclusion and Future Work

- AI-readiness is mostly metadata hygiene.
- Curate content first; format is a second-order choice.
- Adding a metadata summary to a record can improve retrieval efficiency.
- Expand the study with more SME queries and larger data pool.

QR Code

Mentimeter Q&A

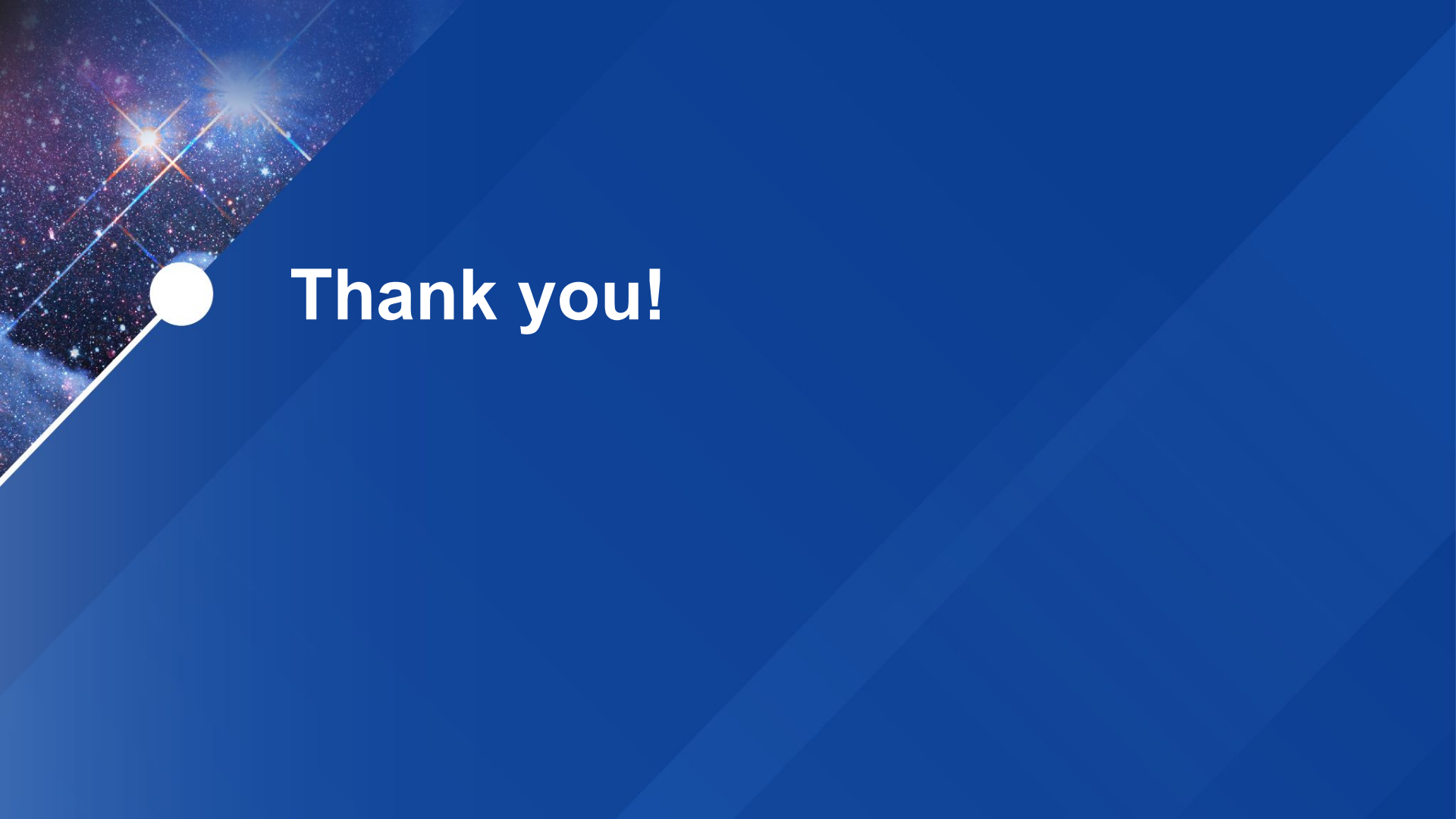
Join at menti.com | use code XXXX XXXX

Call Out or Call to Action

Name

Title



The background is a deep blue gradient with diagonal lines. In the top-left corner, there is a cosmic scene with stars and a bright light source. A white circle is positioned on the left side, connected by a white line to the cosmic scene.

Thank you!